# Supplementary material for "An Infrastructure-Constrained, Observation-Evaluated Inventory of Oil and Gas Fugitive Emissions in Pakistan"

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**Release provenance.** This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

| | |
|---|---|
| Release version | `1.0.0rc69` |
| Commit | `d8d874622f9a` |
| Reporting period | July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30) |
| Reference | S_TAN (household survey; TAN nitrogen) |
| FY2023–24 total | 444.61 Mt $CO_2$e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN |
| Release manifest | `ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16) |
| Source manifest | `c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16) |
| Data-description article | `paper_national/` (ESSD, in preparation), same release |

## 1. Observation-to-field match audit

Table S1 exposes every join used to connect a UNEP MARS source to the nearest World Bank flare and then to the canonical official field ledger. "High" requires an official field, distance no greater than 10 km, a known operator and a non-transmission source. Records at 10–25 km, transmission sources or records without operator metadata require review. A record is unmatched when the named flare field is absent from the 73-field official supply ledger or the distance exceeds 25 km. Unmatched observations remain in the source archive but do not enter named-field statistics. Coordinates, operator and flare identifier are retained in `outputs/fugitives_observation_match_audit_2024.csv`.

Table S1: Nearest-flare portion of the MARS match audit. Source identifiers have the common `PAK_S_` prefix.

| ID | MARS source type | Flare field | Flare operator | km |
|---|---|---|---|---|
| 002 | transmission compressor | Bhit | Prime International Oil & Gas | 0.07 |
| 003 | transmission compressor | Naimat West | United Energy | 0.35 |
| 013 | gas disposal | Loti | OGDC | 3.97 |
| 011 | gathering/boosting | Sharf | Pakistan Petroleum | 0.13 |
| 014 | gathering/boosting | Bhit | Prime International Oil & Gas | 1.20 |
| 015 | O&G facility | Mari | Mari Petroleum Company | 6.56 |
| 016 | O&G facility | Pasakhi | OGDC | 1.98 |
| 005 | gas disposal | Loti | OGDC | 3.99 |
| 006 | gas disposal | Loti | OGDC | 3.64 |
| 001 | gathering/boosting | Mari | Mari Petroleum Company | 0.07 |
| 010 | gas-processing unit | Rajani | United Energy | 0.22 |
| 009 | flare | Sharf | Pakistan Petroleum | 0.06 |
| 008 | gathering/boosting | Qadirpur | OGDC | 0.52 |
| 020 | O&G source | Kadanwari | Prime International Oil & Gas | 0.12 |
| 022 | transmission station | Qadirpur | OGDC | 2.69 |
| 023 | gathering/boosting | Uch | OGDC | 0.27 |
| 027 | O&G facility | Bhit | Prime International Oil & Gas | 14.06 |
| 024 | gathering/boosting | Chanda | OGDC | 0.20 |
| 026 | gathering/boosting | Nashpa | OGDC | 0.35 |
| 025 | gathering/boosting | Manzalai–Kohat (2) | unreported | 0.36 |

Table S2: Canonical-field decision and confidence audit.

| ID | Canonical field | Method | Status | Reason |
|---|---|---|---|---|
| 002 | bhit | exact normalized | review | transmission source |
| 003 | naimat west | exact normalized | unmatched | absent official field; transmission |
| 013 | loti | exact normalized | unmatched | absent official field |
| 011 | sharf | exact normalized | unmatched | absent official field |
| 014 | bhit | exact normalized | high | — |
| 015 | mari | exact normalized | high | — |
| 016 | pasakhi | exact normalized | unmatched | absent official field |
| 005 | loti | exact normalized | unmatched | absent official field |
| 006 | loti | exact normalized | unmatched | absent official field |
| 001 | mari | exact normalized | high | — |
| 010 | rajani | exact normalized | unmatched | absent official field |
| 009 | sharf | exact normalized | unmatched | absent official field |
| 008 | qadirpur | exact normalized | high | — |
| 020 | kadanwari | exact normalized | high | — |
| 022 | qadirpur | exact normalized | review | transmission source |
| 023 | uch | exact normalized | high | — |
| 027 | bhit | exact normalized | review | distance >10 km |
| 024 | chanda | exact normalized | high | — |
| 026 | nashpa | exact normalized | high | — |
| 025 | manzalai | alias | review | operator missing |

## 2. Sparse-field permutation evaluation

All 73 official fields and all zero-observation fields are retained. For each comparison, 10,000 deterministic seeded permutations shuffle observation values either across the whole field set or only among fields in the same province. Spearman tests are two-sided; top-10 overlap tests use the upper tail. The +1 correction is applied to numerator and denominator.

| Observation | Null | Nonzero fields | $\rho$ | $p_\rho$ | Top-10 overlap | $p_{top}$ |
|---|---|---|---|---|---|---|
| VIIRS flare volume | national | 25 | 0.230 | 0.0514 | 4/10 | 0.0224 |
| VIIRS flare volume | within province | 25 | 0.230 | 0.0369 | 4/10 | 0.0257 |
| MARS sources | national | 8 | 0.382 | 0.0003 | 4/8 | 0.0065 |
| MARS sources | within province | 8 | 0.382 | 0.0013 | 4/8 | 0.0374 |
| MARS plumes | national | 8 | 0.382 | 0.0004 | 4/8 | 0.0093 |
| MARS plumes | within province | 8 | 0.382 | 0.0007 | 4/8 | 0.0370 |

Table S3: Seeded permutation diagnostics. Observation quantities are not annualized emissions.

## 3. Completeness and materiality

The screen threshold is 1% of the reference configuration (0.235 Mt $CO_2$e). No numerical upper bound is assigned where a defensible public activity is absent. “IE” denotes inclusion elsewhere in an inclusive factor; “NE” denotes not estimated; “NO” denotes reported non-occurrence; and “NA” denotes no applicable GHG factor in the 2019 Tier-1 table.

| IPCC | Subsegment | Key | Materiality | Priority evidence |
|---|---|---|---|---|
| 1.B.1 | Coal mining | NE | out of scope/unknown | Coal output by mine type and mine-gas content |
| 1.B.2.a.i | Oil exploration | NE | unknown | Exploration wells and testing events |
| 1.B.2.a.ii | Oil production/upgrading | IE | estimated/included | Onshore/offshore split, active wells and practices |
| 1.B.2.a.iii | Oil transport | E/partial | estimated/included | Mode-specific pipeline, truck, rail and tanker throughput |
| 1.B.2.a.iv | Oil refining | E/proxy | estimated/included | Crude feed by refinery |
| 1.B.2.a.v | Oil-product distribution | NA | not applicable | Add NMVOC to the pollutant inventory if required |
| 1.B.2.a.vi | Oil other | NE | not applicable | Incident and spill records |
| 1.B.2.a.vii | Abandoned oil wells | NE | unknown | Plugged/unplugged well census |
| 1.B.2.b.i | Gas exploration | NE | unknown | Exploration wells and testing events |
| 1.B.2.b.ii | Gas production/gathering | IE | estimated/included | Well counts, field throughput and practices |
| 1.B.2.b.iii | Gas processing | E/proxy | estimated/included | Plant throughput, acid-gas removal and compressor seals |
| 1.B.2.b.iv | Gas transmission | E | estimated/included | Independent pipes, LDAR and dry/wet seal shares |
| 1.B.2.b.iv | Gas storage | NO | probably minor/NO | Reassess when an operating facility is reported |
| 1.B.2.b.v | Gas distribution | E | estimated/included | Pipe material, pressure and LDAR coverage |
| 1.B.2.b.vi | Post-meter appliances | E/proxy | estimated/included | Appliance census and commercial multiplier |
| 1.B.2.b.vi | Natural-gas vehicles | NE | unknown | Active CNG-vehicle count by technology |
| 1.B.2.b.vi | Industrial/power post-meter | E | estimated/included | Facility-specific post-meter measurements |
| 1.B.2.b.vii | Other/accidents | NE | potentially material | Rupture, blowdown and super-emitter records |
| 1.B.2.c | Venting/flaring | IE/memo | estimated/included | Operator vented/flared gas and Annex 4A.2 split |

Table S4: Complete notation-key and materiality matrix.

## 4. Material uncertainty-input register

All entries use bounded split-triangular sampling in the reference analysis; the deterministic value is the mode and median. Published IPCC bounds are treated as expert/table ranges, not as fitted confidence intervals. Shared activity-source deviates are applied once across connected segments, while additional proxy-coverage residuals remain independent. The complete register, including minor oil, LNG, appliance, $CO_2$ and $N_2O$ rows and zero-bound handling, is archived in `outputs/fugitives_uncertainty_inputs_2024.csv`.

| Input/group | Central | Lower | Upper | Unit | Shared with | Source |
|---|---|---|---|---|---|---|
| Gas production activity | 32309 | 30694 | 33925 | $MMm^3$ | production/gather/process | Economic Survey |
| Gas-processing activity | 32309 | 25848 | 38771 | $MMm^3$ | production + proxy residual | activity proxy |
| Sour-processing activity | 10339 | 8271 | 12407 | $MMm^3$ | production + proxy residual | 32% default/proxy |
| Distribution pipeline | 201230 | 197205 | 205255 | km | distribution gases | OGRA |
| Post-meter activity | 18685 | 17750 | 19619 | $MMm^3$ | post-meter gases | OGRA end uses |
| Gas-production $CH_4$ EF | 4.09 | 3.272 | 4.908 | $t/MMm^3$ | production gases | IPCC 4.2.4g |
| Gas-processing $CH_4$ EF | 1.83 | 1.647 | 2.013 | $t/MMm^3$ | processing gases | IPCC 4.2.4h |
| Sour-processing $CO_2$ EF | 66.6 | 0 | 133.2 | $t/MMm^3$ | sour-processing gases | IPCC 4.2.4h |
| Distribution $CH_4$ EF | 1.17 | 0.936 | 2.574 | t/km | distribution gases | IPCC 4.2.4j |
| Post-meter $CH_4$ EF | 6.1 | 2.44 | 9.76 | $t/MMm^3$ | post-meter gases | IPCC 4.2.4k |

Table S5: Material reference-configuration inputs.

## 5. Conditional uncertainty and distribution shape

For reported value $x$, proportional lower and upper ranges $l$ and $h$, and $u \sim \mathcal{U}(0,1)$, the bounded split-triangular multiplier is

$$m(u) = \begin{cases} 1 - l + l\sqrt{2u}, & u < 0.5, \\ 1 + h - h\sqrt{2(1-u)}, & u \geq 0.5. \end{cases} \tag{1}$$

The reported input is the mode and median. Gas production, gathering and processing share the production-activity draw; additional processing-coverage residuals are independent. Oil production and its transport fallback share the oil draw. Gases belonging to one physical factor group share a quantile transformed through their gas-specific bounds. All simulations use 10,000 iterations and seed 42.

| Experiment | Distribution | Deterministic | P2.5 | Median | P97.5 |
|---|---|---|---|---|---|
| Reference | split triangular | 23.499 | 21.277 | 24.141 | 30.537 |
| Central (geometric-mean practice) | split triangular | 17.678 | 15.798 | 18.137 | 21.309 |
| Lower practices | split triangular | 14.894 | 13.113 | 15.144 | 17.311 |
| No industrial post-meter | split triangular | 20.079 | 18.489 | 20.526 | 26.836 |
| Minimum coverage | split triangular | 9.601 | 8.864 | 9.745 | 10.692 |
| Reference shape test | bounded uniform | 23.499 | 21.351 | 27.140 | 33.140 |
| Reference shape test | bounded log-uniform[a] | 23.499 | 20.300 | 25.239 | 31.909 |

Table S6: Conditional 95% intervals for the five interpretable configurations, and distribution-shape sensitivity, Mt $CO_2e$. The central member is a synthesized geometric-mean reading, priced and never selected.

[a]Inputs with a zero lower bound use bounded uniform sampling.

## 6. Official-inventory and computational audit

The CRT audit preserves the source ZIP and workbook hashes, worksheet names, cell coordinates, raw values, units, formulas and notation keys. Generated tables reproduce each numeric activity–factor row in native gases, convert the submitted total to the common GWP basis and close every step of the CRT-to-reconstruction waterfall. The complete records are archived in `outputs/fugitives_unfccc_crt_cell_audit.csv`, `outputs/fugitives_unfccc_crt_reproduction.csv` and `outputs/fugitives_unfccc_crt_waterfall_2022.csv`.

Automated QA covers factor units, registered activity years, source-row and segment closure, pipeline and gas-flow identities, all 280 structural configurations, conditional interval ordering, field-supply reconciliation, observation-match completeness, permutation reproducibility, provincial reconciliation and manuscript-output agreement. The run manifest and machine-readable result are stored in `outputs/fugitives_run_manifest_2024.json` and `outputs/fugitives_qa_2024.csv`.